Lecture 1: SPDE — Introduction

Tien Anh Vu

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Introduction

Example: Dean–Kawasaki equation

This is a motivating example for SPDEs arising from interacting particle systems.
(Dean 1996, Kawasaki 1998)

Consider independent Brownian motions $B^1, B^2, B^3, \ldots$ (independent BMs), and a smooth function $f \in C_b^2(\mathbb{R})$, i.e., twice differentiable with bounded derivatives and bounded values.

Empirical average. The empirical measure of N particles, each performing independent BM starting from x_0, is averaged against f.

Then for N particles:

$$\eta_t^N(f)(\omega) \;:=\; \frac{1}{N}\sum_{i=1}^{N} f\big(B_t^i + x_0\big)(\omega)$$

This is the empirical (sample) average of f evaluated at the particle positions $B_t^i + x_0$.

Remark 1. We can rewrite this as an integral against the empirical measure:

$$\eta_t^N(f)(\omega) \;=\; \int f\, d\left(\frac{1}{N}\sum_{i=1}^{N} \delta_{B_t^i + x_0}\right)(\omega).$$

Law of large numbers. By the SLLN, the empirical average converges P-a.s. as $N \to \infty$.

By the **strong law of large numbers (SLLN)**, as $N \to \infty$:

$$\eta_t^N(f) \;\xrightarrow{P\text{-a.s.}}\; E\big(f(B_t + x_0)\big).$$

The distribution of $B_t + x_0$ is $\mathcal{N}(x_0, t)$, so the expectation is an integral against the Gaussian density.

Since $B_t + x_0 \sim \mathcal{N}(x_0,\, t)$, the right-hand side equals:

$$E\big(f(B_t + x_0)\big) \;=\; \int_{\mathbb{R}} f(x)\, p(t,x)\, dx,$$

where $p(t,x)$ is the **Gaussian heat kernel**:

$$p(t,x) \;=\; \frac{1}{\sqrt{2\pi t}}\, e^{-\frac{(x-x_0)^2}{2t}}.$$

Remark 2. The convergence $\eta_t^N(f) \xrightarrow{P\text{-a.s.}} E\big(f(B_t + x_0)\big)$ holds for each fixed t and each test function f. The right-hand side is deterministic: all randomness has been averaged out. The Gaussian density $p(t,x)$ arises because the sum of N i.i.d. displacements (each distributed as B_t) concentrates around the common mean, and each particle's position has the distribution $\mathcal{N}(x_0, t)$.

Connection to the heat equation

The density $p(t,x)$ solves the heat equation. This is the classical connection between Brownian motion and parabolic PDEs.

Then $p(t,x)$ solves the following **PDE** (heat equation):

$$\boxed{\begin{cases} \partial_t\, p(t,x) \;=\; \frac{1}{2}\,\partial_{xx}\, p(t,x), & t>0,\ x\in\mathbb{R}, \\[2ex] p(t,x)\,dx \;\longrightarrow\; \delta_{x_0}(dx), & \text{(initial condition).} \end{cases}}$$

Remark 3. This is the **heat equation**. The initial condition says that as $t \to 0^+$, the measure $p(t,x)\,dx$ converges (in the sense of distributions) to the Dirac delta δ_{x_0}. Physically: heat concentrated at x_0 at time 0 diffuses according to the heat equation. The solution $p(t,x)$ is the fundamental solution (Green's function) of the heat operator.

Remark 4. Intuitively, the heat equation describes diffusion: an initial concentration of "heat" at x_0 spreads out over time, with the profile given by the Gaussian $p(t,x)$. The solution has **negative curvature** — the temperature decreases over time at the peak and spreads to the sides.

From microscopic to macroscopic description

Brownian motion provides the **microscopic description** (individual particles). The PDE (heat equation) provides the **macroscopic description** (density of particles).

Remark 5. This passage from *microscopic* to *macroscopic* is a central theme in mathematical physics and probability theory. At the microscopic level, we track each particle individually (a system of N SDEs); at the macroscopic level, we describe the *collective behaviour* via a single PDE for the density. The law of large numbers is the mathematical mechanism that justifies this passage: random fluctuations average out as $N \to \infty$, and only the deterministic drift survives. This is sometimes called a **mean-field limit** or **hydrodynamic limit**.

Fluctuations around LLN — Itô's formula

Apply Itô's formula to the empirical average:

$$\begin{aligned} d\left(\frac{1}{N}\sum_{i=1}^{N} f\big(B_t^i + x_0\big)\right) \;&=\; \frac{1}{N}\sum_{i=1}^{N} df\big(B_t^i + x_0\big) \\ &=\; \underbrace{\frac{1}{N}\sum_{i=1}^{N} f'\big(B_t^i + x_0\big)\,dB_t^i}_{\text{Itô integral}} \;+\; \underbrace{\frac{1}{N}\sum_{i=1}^{N} \frac{1}{2}\,f''\big(B_t^i + x_0\big)\,dt}_{\text{correction term}}. \end{aligned}$$

We can identify the two terms:

- The Itô integral part defines a **local martingale** $M_t^N(f)$.
- The correction term equals $\eta_t^N\big(\frac{1}{2}f''\big)\,dt$.

So in shorthand:

$$d\,\eta_t^N(f) \;=\; dM_t^N(f) \;+\; \eta_t^N\big(\tfrac{1}{2}f''\big)\,dt.$$

This is the **asymptotic of the Itô integral**: as $N \to \infty$, $M_t^N(f) \to 0$, and we recover the PDE.

Remark 6. The decomposition $d\,\eta_t^N(f) = dM_t^N(f) + \eta_t^N(\frac{1}{2}f'')\,dt$ is a **semimartingale decomposition**: every semimartingale can be written as a local martingale plus a finite-variation (drift) part. The martingale part $M_t^N(f)$ captures the *randomness* (fluctuations due to the individual Brownian motions), while the drift part $\eta_t^N(\frac{1}{2}f'')\,dt$ captures the *deterministic trend* (the diffusion described by the heat equation). As $N \to \infty$, the noise averages out and only the drift remains — this is the law of large numbers at the level of processes.

Remark 7. The factor $\frac{1}{2}f''$ appearing in the drift is the **generator of Brownian motion**. In general, if X_t is a diffusion with generator $\mathcal{A}$, then Itô's formula gives $d\,f(X_t) = \mathcal{A}f(X_t)\,dt +$ (martingale). For standard BM, $\mathcal{A} = \frac{1}{2}\partial_{xx}$, hence the $\frac{1}{2}f''$ term.

Quadratic variation. The quadratic variation is computed using the **Itô isometry**:

$$\begin{aligned} E\Big(\big(M_t^N(f)\big)^2\Big) &= \frac{1}{N^2}\sum_{i=1}^N E\bigg(\Big(\int_0^t f'\big(B_s^i + x_0\big)\,dB_s^i\Big)^2\bigg) \\ &= \frac{1}{N^2}\sum_{i=1}^N E\bigg(\int_0^t f'\big(B_s^i + x_0\big)^2\,ds\bigg) \\ &= \frac{1}{N}\,E\bigg(\int_0^t f'\big(B_s + x_0\big)^2\,ds\bigg). \end{aligned}$$

In the last step, we used that the B^i are i.i.d., so all N terms are equal, giving a factor $1/N$.

Quadratic variation process. We have

$$\big\langle M^N(f)\big\rangle_t = \frac{1}{N^2}\sum_{i=1}^N \int_0^t f'\big(B_s^i + x_0\big)^2\,ds = \frac{1}{N}\int_0^t \eta_s^N\big((f')^2\big)\,ds = O\Big(\frac{1}{N}\Big).$$

The quadratic variation is $O(1/N)$, confirming that the martingale part vanishes as $N \to \infty$. The fluctuations around the deterministic limit are of order $1/\sqrt{N}$.

Remark 8. The **Itô isometry** states that for a (predictable) integrand ϕ_s and a Brownian motion B_s:

$$E\bigg[\Big(\int_0^t \phi_s\,dB_s\Big)^2\bigg] = E\bigg[\int_0^t \phi_s^2\,ds\bigg].$$

It converts a stochastic quantity (second moment of the Itô integral) into a deterministic one (expected integral of the integrand squared). This is why the cross terms $i \neq j$ vanish in the computation: the Brownian motions B^i and B^j are *independent*, so $E\big(\int \phi\,dB^i \cdot \int \psi\,dB^j\big) = 0$.

Remark 9. The quadratic variation being $O(1/N)$ means that $E\big((M_t^N)^2\big) \to 0$, which implies $M_t^N \to 0$ in $L^2(\Omega)$ (and hence in probability). This is a *quantitative* version of the law of large numbers: the variance of the empirical average decreases at rate $1/N$ (equivalently, the standard deviation decreases at rate $1/\sqrt{N}$).

Integral form

Recall from $(\star)$:

$$d\,\eta_t^N(f) = dM_t^N(f) + \eta_t^N\big(\tfrac{1}{2}f''\big)\,dt.$$

As $N \to \infty$, the martingale part $M_t^N(f) \to 0$ (since its quadratic variation is $O(1/N)$), so only the drift term survives. Integrating in time:

$$\eta_t^N(f) \;=\; \eta_0^N(f) \;+\; \int_0^t \eta_s^N\big(\tfrac{1}{2}f''\big)\,ds.$$

In the limit $N \to \infty$, this becomes the **weak (distributional) form** of the heat equation: for every test function $f \in C_b^2(\mathbb{R})$, the limiting density η_t satisfies $\eta_t(f) = \eta_0(f) + \int_0^t \eta_s(\frac{1}{2}f'')\,ds$.

Remark 10. The integral form $\eta_t^N(f) = \eta_0^N(f) + M_t^N(f) + \int_0^t \eta_s^N(\frac{1}{2}f'')\,ds$ is a **mild formulation**: rather than requiring differentiability of η_t^N in time, the equation is expressed as an integral identity. This is advantageous because the empirical measure η_t^N is a random sum of Dirac masses, which is not differentiable in the classical sense. The passage to the limit $N \to \infty$ is performed at the level of this integral identity, which is natural for probabilistic arguments where limits are taken in distribution or in L^p, rather than pointwise for derivatives.

Density form. The empirical measure η_s^N is a sum of Dirac masses. Its density (in the distributional sense) is: Define the empirical density:

$$\eta_s^N(x) \;:=\; \frac{1}{N}\sum_{i=1}^N \delta_{B_s^i + x_0}(x).$$

Assuming we can pass to the limit (i.e., replace η_s^N by a smooth density η_s), we use **integration by parts** (twice) to move the derivatives from f onto η:

$$\eta_t^N\big(\tfrac{1}{2}f''\big) \;=\; \int f''(x)\;\tfrac{1}{2}\,\eta_t^N(x)\,dx \;=\; \int f(x)\;\tfrac{1}{2}\,\partial_{xx}\,\eta_t^N(x)\,dx,$$

where the boundary terms vanish because $f \in C_b^2$. This gives

$$\eta_t^N\big(\tfrac{1}{2}f''\big)\,dt \;=\; \tfrac{1}{2}\,\partial_{xx}\,\eta_t^N(f)\,dt.$$

This is precisely the **weak form of the heat equation**: the identity $\int f \cdot \frac{1}{2}f''\,\eta\,dx = \int f \cdot \frac{1}{2}\partial_{xx}\eta\,dx$ holds after integration by parts, confirming that the limiting density solves $\partial_t \eta = \frac{1}{2}\partial_{xx}\eta$.

Remark 11. **Integration by parts** is the key tool that converts the *weak formulation* (derivatives on the test function f) into the *strong formulation* (derivatives on the density η). In the weak form, one only requires f to be smooth; the density η can be rough. This is essential because the empirical measure η_s^N is a sum of Dirac masses (not differentiable), so the weak formulation is the natural one at finite N. Only in the limit $N \to \infty$, when η becomes smooth, can we pass to the strong (classical) PDE form.

Remark 12. The weak form $\eta_t(f) = \eta_0(f) + \int_0^t \eta_s(\frac{1}{2}f'')\,ds$ for all test functions f is the **mild/weak solution concept** for the heat equation. It says: "test the density against any smooth function f and check that the resulting real-valued function of time satisfies an ODE." This avoids requiring pointwise differentiability of η and is the standard notion of solution for PDEs with irregular data.

Fluctuations around SLLN

While the martingale $M_t^N(f)$ vanishes in the limit, its **fluctuations** (after rescaling by $\sqrt{N}$) converge to a non-trivial limit. The martingale part is:

$$M_t^N(f) \;=\; \frac{1}{N}\sum_{i=1}^N \int_0^t f'\big(B_s^i + x_0\big)\,dB_s^i.$$

We can rewrite this by splitting $\frac{1}{N} = \frac{1}{\sqrt{N}} \cdot \frac{1}{\sqrt{N}}$:

$$M_t^N(f) \;=\; \frac{1}{\sqrt{N}} \cdot \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \int_0^t f'\big(B_s^i + x_0\big)\, dB_s^i.$$

Remark 13. The splitting $\frac{1}{N} = \frac{1}{\sqrt{N}} \cdot \frac{1}{\sqrt{N}}$ is the key step that reveals the **central limit theorem scaling**. The inner factor $\frac{1}{\sqrt{N}} \sum_{i=1}^{N}$ is the CLT-normalised sum, which converges to a non-trivial Gaussian limit. The outer factor $\frac{1}{\sqrt{N}}$ then sends the whole expression to zero, consistently with the LLN. Removing the outer $\frac{1}{\sqrt{N}}$ (i.e., studying $\sqrt{N}\, M_t^N$) isolates the fluctuations.

The key observation is that the 1st and 2nd moments of the rescaled sum coincide in the limit (by the law of large numbers for the quadratic variation).

Limiting density. In the limit, $\eta_t(x)\, dx = p(t,x)\, dx$ (the Gaussian heat kernel), and the martingale part, after rescaling by $\sqrt{N}$, converges to a **stochastic integral** involving the square root of the limiting density:

$$M_t^N(f) \;\approx\; \frac{1}{\sqrt{N}} \int_0^t \!\!\int f'(x) \sqrt{\eta_s^N(x)}\, d\xi(s,x),$$

where $d\xi(s,x)$ denotes **space-time white noise** (informally: one independent Brownian motion per spatial location x).

The appearance of $\sqrt{\eta_s^N(x)}$ is natural: each particle at position x contributes an independent Brownian increment, and the "density of particles" near x is $\eta_s^N(x)$, so by the central limit theorem the fluctuations scale like $\sqrt{\eta_s^N(x)}$.

Remark 14. The factor $\sqrt{\eta_s^N(x)}$ can be understood from two complementary viewpoints. **Probabilistically:** Near position x, there are approximately $N\, \eta_s^N(x)\, \Delta x$ particles in a small interval of length Δx. Each contributes an independent noise of order 1, so by the CLT the total fluctuation is of order $\sqrt{N\, \eta_s^N(x)\, \Delta x}$. After dividing by N (to form the empirical density), the noise per unit length is proportional to $\sqrt{\eta_s^N(x)/N}$. **Analytically:** The $\sqrt{\eta}$ factor is dictated by the *quadratic variation*: the Itô isometry forces the noise coefficient to be the square root of the integrand appearing in the quadratic variation $\langle M^N(f)\rangle_t$, which is $\eta_s^N\big((f')^2\big)$.

The quadratic variation confirms this:

$$\big\langle M^N(f)\big\rangle_t \;=\; \frac{1}{N} \int_0^t \!\!\int \big(f'(x)\big)^2\, \eta_s^N(x)\, dx\, ds,$$

which is exactly the Itô isometry applied to the stochastic integral representation above.

CLT for the rescaled martingale

The rescaled martingale converges:

$$\sqrt{N}\, M_t^N(f) \;\longrightarrow\; \int_0^t \sqrt{\eta_s\big((f')^2\big)}\, dB_s \qquad \text{(Itô integral w.r.t. BM).}$$

For each test function f, the limit is Gaussian:

$$f \;\longmapsto\; \mathcal{N}\!\left(0,\; \int_0^t \eta_s\big((f')^2\big)\, ds\right).$$

Remark 15. This is the **martingale central limit theorem** (martingale CLT) in action. The classical CLT says that $\frac{1}{\sqrt{N}}\sum_{i=1}^{N} X_i \to \mathcal{N}(0,\sigma^2)$ for i.i.d. random variables. Here, the summands are *stochastic integrals* $\int_0^t f'(B_s^i+x_0)\,dB_s^i$, which are themselves random variables (in fact, martingales). The martingale CLT guarantees that their normalised sum converges to a Gaussian, with variance determined by the **predictable quadratic variation**.

Remark 16. The variance $\int_0^t \eta_s\big((f')^2\big)\,ds$ of the limiting Gaussian depends on the *test function* f through $(f')^2$ and on the *limiting density* η_s. This means that the fluctuations are spatially *heterogeneous*: regions with higher particle density $\eta_s(x)$ contribute more variance. This is the infinite-dimensional analogue of the classical result that the variance of a sample mean depends on the population variance.

Leading to the SDE. Combining the drift and the martingale part (using the Itô isometry):

$$d\,\eta_t^N(f) \;\approx\; \eta_t^N\big(\tfrac{1}{2}f''\big)\,dt \;+\; \frac{1}{\sqrt{N}}\,\sqrt{\eta_t^N\big((f')^2\big)}\;dB_t.$$

Remark 17. This approximate SDE shows the two competing effects:

- The **drift** $\eta_t^N(\frac{1}{2}f'')\,dt$ drives the system towards the deterministic heat equation.
- The **noise** of order $1/\sqrt{N}$ captures the random fluctuations due to finite particle number.

As $N\to\infty$, the noise term vanishes and we recover the deterministic PDE. For large but finite N, this equation quantifies how the empirical measure deviates from the hydrodynamic limit.

Space-time white noise representation

For $s>0$, we can write $\eta_s^N(dx)=\eta_s^N(x)\,dx$, and using **space-time white noise** $\xi(t,x)$, i.e.

$$\mathbb{E}\big(\xi(t,x)\,\xi(s,y)\big) \;=\; \delta(t-s)\,\delta(x-y) \qquad \text{(uncorrelated)},$$

we can write the following chain of equalities.

The L^2 norm squared of the martingale (i.e. the quadratic variation) can be rewritten as:

$$\begin{aligned}\frac{1}{N}\,\mathbb{E}\bigg(\int_0^t \eta_s^N\big((f')^2\big)\,ds\bigg) \;&=\; \frac{1}{N}\,\mathbb{E}\bigg(\int_0^t\!\!\int_{\mathbb{R}} (f')^2(x)\;\eta_s^N(x)\,dx\,ds\bigg)\\ &=\; \frac{1}{N}\,\mathbb{E}\bigg[\bigg(\int_0^t\!\!\int_{\mathbb{R}} f'(x)\;\sqrt{\eta_s^N(x)}\;d\xi(s,x)\bigg)^2\bigg].\end{aligned}$$

The last equality follows from the **Itô isometry** for the stochastic integral against space-time white noise: if $g(s,x)=f'(x)\,\sqrt{\eta_s^N(x)}$, then $g^2=(f')^2(x)\,\eta_s^N(x)$, confirming the identity.

Remark 18. **Space-time white noise** $\xi(t,x)$ is the infinite-dimensional analogue of standard white noise. Formally, ξ is a generalised Gaussian random field indexed by $(t,x)\in[0,T]\times\mathbb{R}$ with covariance $E\big(\xi(t,x)\,\xi(s,y)\big)=\delta(t-s)\,\delta(x-y)$. Rigorously, it is defined as a random distribution: for each test function $\varphi\in L^2([0,T]\times\mathbb{R})$, the pairing $\xi(\varphi):=\iint\varphi(t,x)\,\xi(t,x)\,dt\,dx$ is a centred Gaussian random variable with variance $\|\varphi\|_{L^2}^2$. This is the natural noise driving SPDEs, just as standard BM is the natural noise driving SDEs.

Remark 19. The passage from the sum of N independent martingales to a single stochastic integral against space-time white noise is the key conceptual step. Heuristically, each particle at position x contributes an independent "Brownian kick"; as $N\to\infty$, these individual kicks merge into a continuum of independent noises parametrised by x, which is precisely what space-time white noise models.

The Dean–Kawasaki SPDE

This leads to the **Dean–Kawasaki SPDE**:

$$\boxed{\partial_t \eta_t^N \;=\; \tfrac{1}{2}\,\partial_{xx}\,\eta_t^N(x) \;+\; \frac{1}{\sqrt{N}}\,\partial_x\Big(\sqrt{\eta_t^N(x)}\;\xi(t,x)\Big)\,.}$$

Remark 20. This SPDE is **ill-posed**: no general (function-valued) solution exists. The noise term involves both a square root of the solution and a spatial derivative of the space-time white noise, creating severe irregularity.

Remark 21. The Dean–Kawasaki SPDE has the structure of a **stochastic conservation law**: the noise enters through a spatial derivative $\partial_x(\cdots)$, which ensures that the total mass $\int \eta_t^N(x)\,dx$ is formally conserved (the noise only *redistributes* mass, it does not create or destroy it). This is physically natural since the number of particles is fixed at N.

Remark 22. Despite its ill-posedness as a classical SPDE, the Dean–Kawasaki equation is extremely useful as a **formal identity**: it correctly encodes both the law-of-large-numbers limit (heat equation) and the fluctuations (the $1/\sqrt{N}$ noise). Recent mathematical work has given rigorous meaning to this equation using **measure-valued solutions** or **renormalisation** techniques.

OU process. As a simpler model, the **Ornstein–Uhlenbeck (OU) process** is the solution of the linear SDE:

$$dX_t \;=\; A\,X_t\,dt \;+\; C\,dW_t.$$

Remark 23. The OU process is the simplest non-trivial example of a **linear SDE**. Compared to the Dean–Kawasaki equation, it is well-posed and explicitly solvable:

$$X_t \;=\; e^{At}\,X_0 \;+\; \int_0^t e^{A(t-s)}\,C\;dW_s.$$

Here A is a (possibly unbounded) linear operator generating mean reversion (if $A < 0$), and C controls the noise intensity. The OU process will serve as the **linear prototype** for the general theory of SPDEs on Hilbert spaces that follows in Chapter 1.

Remark 24. In finite dimensions ($H = \mathbb{R}^d$), the OU process has a Gaussian invariant measure $\mathcal{N}(0, Q_\infty)$ provided A is stable (all eigenvalues have negative real part), where Q_∞ solves the **Lyapunov equation** $AQ_\infty + Q_\infty A^* + CC^* = 0$. In infinite dimensions, the analogous condition requires Q_∞ to be trace-class, which imposes a *compatibility condition* between the operators A and C.

Chapter 1: Stochastic Analysis on Hilbert Spaces

Hilbert spaces

Recall that $\big(L^2(\mu),\, \langle\cdot,\cdot\rangle\big)$ is a **complete normed vector space** (Hilbert space).

The inner product is given by,

$$\langle f, g\rangle = \int f\, g\, d\mu.$$

A linear operator between Hilbert spaces $U \to V$ is **continuous** if and only if its operator norm is finite:

$$\sup_{\|u\|_U \le 1} \|L\, u\|_V \;<\; \infty.$$

Remark 25. A **Hilbert space** is the natural infinite-dimensional generalisation of Euclidean space $\mathbb{R}^d$: it is a (possibly infinite-dimensional) vector space equipped with an inner product that induces a complete norm. Completeness means that every Cauchy sequence converges, which is essential for taking limits (e.g., of partial sums in Fourier series or of approximating solutions to equations).

Remark 26. The equivalence "continuous $\Leftrightarrow$ bounded" for linear operators is a consequence of linearity: a linear map that is continuous at one point is automatically continuous everywhere. The **operator norm** $\|L\|_{\mathrm{op}} := \sup_{\|u\|\le 1} \|Lu\|$ measures the maximum "stretching factor" of L. The space of all bounded linear operators $H \to H$, denoted $\mathcal{L}(H)$, is itself a Banach space under this norm.

Example: ℓ^2. The space of square-summable sequences:

$$\ell^2 \;=\; \left\{ (u_n)_{n\in\mathbb{N}} \in \mathbb{R}^{\mathbb{N}} \;:\; \sum_{n=1}^{\infty} u_n^2 \;<\; \infty \right\}.$$

The **inner product** on ℓ^2 is:

$$\langle u, v\rangle \;=\; \sum_{k=1}^{\infty} u_k\, v_k.$$

Remark 27. The space ℓ^2 is the **prototypical separable Hilbert space**. By Parseval's identity, every separable Hilbert space is isometrically isomorphic to ℓ^2: choosing a CONS $\{e_k\}$ and mapping $x \mapsto (\langle x, e_k\rangle)_k$ gives a bijective isometry $H \cong \ell^2$. Thus, working in ℓ^2 is essentially the same as working in any other separable Hilbert space. The standard basis of ℓ^2 is $e_k = (0, \ldots, 0, 1, 0, \ldots)$ with a 1 in position k.

Separability. A Hilbert space is called **separable** if there exists a **countable dense subset** $H_0 \subset H$.

Remark 28. Separability is a mild assumption satisfied by all Hilbert spaces that arise in practice (e.g., $L^2(M)$ for σ-finite measure spaces M, and ℓ^2). Its key consequence is the existence of a **countable** complete orthonormal system (CONS) $\{e_k\}_{k=1}^{\infty}$, which allows us to represent every element $x \in H$ as a convergent series $x = \sum_k \langle x, e_k\rangle\, e_k$. Without separability, the CONS could be uncountable, and the theory of Gaussian measures and trace-class operators becomes significantly more delicate.

Definition 1.5: Gaussian measure

Definition 1 (Gaussian measure)**.** A probability measure μ on $(H, \mathcal{B}(H))$ is called **Gaussian** if for all $h \in H$, the linear functional

$$\ell_h \colon H \longrightarrow \mathbb{R}, \qquad g \longmapsto \langle g, h \rangle_H,$$

is **normally distributed** under μ. That is, there exist $m(h) \in \mathbb{R}$ and $\sigma^2(h) \in \mathbb{R}^+$ such that for all $t \in \mathbb{R}$:

$$\phi_{\ell_h}(t) := \int e^{it\,\ell_h}\, d\mu \;=\; e^{it\,m(h) - \frac{1}{2}t^2\,\sigma^2(h)}.$$

Remark 29. Any linear functional on a Hilbert space can be represented as a scalar product (inner product) with some element of H. This is the **Riesz representation theorem**.

Remark 30. Definition 1.5 says that a measure μ on H is Gaussian if *every one-dimensional projection* of μ is Gaussian. This is the infinite-dimensional analogue of the characterisation of multivariate Gaussians in $\mathbb{R}^d$: a random vector $X \in \mathbb{R}^d$ is Gaussian if and only if $a^T X$ is (univariate) Gaussian for every $a \in \mathbb{R}^d$. The inner product $\langle g, h \rangle_H$ plays the role of the linear combination $a^T X$.

Linearity properties. Since $\ell_{h_1+h_2} = \ell_{h_1} + \ell_{h_2}$ (the map $h \mapsto \ell_h$ is linear), we have:

$$\int \ell_{h_1+h_2}\, d\mu \;=\; \int \ell_{h_1}\, d\mu \;+\; \int \ell_{h_2}\, d\mu,$$

and therefore the **mean operator** is linear:

$$m(h_1 + h_2) \;=\; m(h_1) \;+\; m(h_2).$$

Remark 31. The variance $\sigma^2(h)$ depends **quadratically** on h, which leads to the covariance operator Q satisfying $\sigma^2(h) = \langle Qh, h \rangle$. More precisely, the quadratic dependence means that $\sigma^2(\alpha h) = \alpha^2 \sigma^2(h)$ for all $\alpha \in \mathbb{R}$, and the map $(h_1, h_2) \mapsto \mathrm{Cov}(\ell_{h_1}, \ell_{h_2})$ is a *bilinear form*. The covariance operator Q is the unique self-adjoint operator that represents this bilinear form via the inner product: $\mathrm{Cov}(\ell_{h_1}, \ell_{h_2}) = \langle Qh_1, h_2 \rangle$.

Theorem 1.6: Characterisation of Gaussian measures

Theorem 1 (Characterisation of Gaussian measures on Hilbert spaces)**.** A probability measure μ on $\big(H, \mathcal{B}(H)\big)$ is **Gaussian** if and only if

$$\hat{\mu}(h) := \int_H e^{i\langle g, h \rangle_H}\, \mu(dg) \;=\; e^{i\langle m, h \rangle_H - \frac{1}{2}\langle Qh, h \rangle_H} \qquad \text{for all } h \in H,$$

where:

a) $m \in H$ is the **mean vector**,

b) $Q \colon H \to H$ is a linear, continuous operator on H that is

- **symmetric** (self-adjoint), i.e. $\langle Qh, g \rangle_H = \langle h, Qg \rangle_H$ for all $h, g \in H$,
- **positive semi-definite**, i.e. $\langle Qh, h \rangle_H \geq 0$ for all $h \in H$,
- of **finite trace**, i.e. $\mathrm{tr}(Q) \;=\; \sum_{k=1}^{\infty} \langle Qe_k, e_k \rangle \;<\; \infty$

for any (hence for all) complete orthonormal system (CONS) $\{e_k\}$ of H.

In this case we write $\mu = \mathcal{N}(m,\, Q)$, and Q is called the **covariance operator** of μ.

Remark 32. The left-hand side of the characterisation formula is the **characteristic function** (Fourier transform) of the measure μ evaluated at h. In finite dimensions, a Gaussian $\mathcal{N}(m, \Sigma)$ on $\mathbb{R}^d$ has characteristic function $\exp\bigl(i\, m^T h - \frac{1}{2}\, h^T \Sigma\, h\bigr)$. Theorem 1.6 is the infinite-dimensional generalisation: the covariance *matrix* Σ is replaced by a covariance *operator* Q, and the finite-dimensional inner product $m^T h$ becomes $\langle m, h\rangle_H$.

Remark 33. The **trace-class** condition $\mathrm{tr}(Q) < \infty$ is essential and has no analogue in finite dimensions (where every symmetric positive semi-definite matrix automatically has finite trace). In infinite dimensions, not every bounded symmetric positive operator is trace-class. The trace-class condition ensures that μ is concentrated on H (rather than on a larger space) and that the Gaussian measure is well-defined as a *Radon* probability measure on H.

Remark 34. The definition of $\mathrm{tr}(Q)$ is **independent of the choice of CONS** $\{e_k\}$. This follows from the fact that for a non-negative self-adjoint operator Q, the quantity $\sum_k \langle Qe_k, e_k\rangle$ is the same for every CONS (a standard result in functional analysis).

Properties of $\mathcal{N}(m,\, Q)$. In particular, the following hold:

(i) **Mean:**

$$\int_H \langle x,\, h\rangle\, \mu(dx) \;=\; \langle m,\, h\rangle \qquad \text{for all } h \in H.$$

(ii) **Covariance:**

$$\int_H \bigl(\langle x,\, h_1\rangle - \langle m,\, h_1\rangle\bigr)\bigl(\langle x,\, h_2\rangle - \langle m,\, h_2\rangle\bigr)\, \mu(dx) \;=\; \langle Qh_1,\, h_2\rangle \;=\; \langle h_1,\, Qh_2\rangle.$$

(iii) **Second moment / trace identity:**

$$\begin{aligned}\int_H \|x-m\|_H^2\, \mu(dx) \;&=\; \int_H \sum_{\ell=1}^{\infty} \langle x-m,\, e_\ell\rangle^2\, \mu(dx)\\ &=\; \sum_{\ell=1}^{\infty} \int_H \langle x-m,\, e_\ell\rangle^2\, \mu(dx)\\ &=\; \sum_{\ell=1}^{\infty} \langle Qe_\ell,\, e_\ell\rangle \;=\; \mathrm{tr}(Q).\end{aligned}$$

Remark 35. Property (i) says that m is the **Pettis integral** (or weak integral) of the identity map under μ: the mean of each "coordinate" $\langle x, h\rangle$ is $\langle m, h\rangle$.

Property (ii) shows that Q encodes all second-order information. Setting $h_1 = h_2 = h$ gives $\mathrm{Var}_\mu\bigl(\langle x, h\rangle\bigr) = \langle Qh, h\rangle$, recovering the variance formula from Definition 1.5.

Property (iii) identifies the **expected squared norm** of the centred random variable $x - m$ with the trace of the covariance operator. The interchange of sum and integral (second equality) is justified by the **monotone convergence theorem**, since $\langle x - m, e_\ell\rangle^2 \geq 0$. This gives a concrete probabilistic interpretation of $\mathrm{tr}(Q)$: it is the total expected "energy" (squared norm) of the fluctuations around the mean, distributed across all modes e_ℓ of the Hilbert space.

Parseval's identity. If $\{e_i\}$ is a CONS of H and $x \in H$, then writing $x_i := \langle x,\, e_i\rangle_H$ we have

$$\|x\|_H^2 \;=\; \sum_{i=1}^{\infty} x_i^2 \;=\; \sum_{i=1}^{\infty} \langle x,\, e_i\rangle^2.$$

Remark 36. Parseval's identity says that every element of a separable Hilbert space is completely determined by its Fourier coefficients $x_i = \langle x, e_i\rangle$, and the norm is computed as the ℓ^2-norm of these coefficients. In other words, the map $x \mapsto (\langle x, e_i\rangle)_{i\geq 1}$ is an **isometric isomorphism** from H to ℓ^2. This identity was used in property (iii) above when we expanded $\|x - m\|_H^2 = \sum_\ell \langle x - m, e_\ell\rangle^2$.

Example: Wiener process as a Gaussian measure on $L^2([0,T])$

Let $(\Omega,\, \mathcal{F},\, P)$ be a probability space carrying a **standard one-dimensional Brownian motion** $(B_t)_{t\in[0,T]}$ with continuous sample paths.

The sample-path mapping. Each realisation of the Brownian motion is a continuous function on $[0,T]$. We define the **sample-path mapping**

$$\beta \,:\, \Omega \;\longrightarrow\; C([0,T]), \qquad \omega \;\longmapsto\; \big(t \mapsto B_t(\omega)\big).$$

Since continuous functions on a compact interval are square-integrable,

$$C([0,T]) \;\subseteq\; L^2([0,T]) \qquad \text{(Hilbert space)},$$

so every sample path $t \mapsto B_t(\omega)$ is an element of $L^2([0,T])$.

Remark 37. The inclusion $C([0,T]) \subseteq L^2([0,T])$ is *continuous* but *not* surjective: L^2 contains equivalence classes of functions that may be highly irregular, whereas sample paths of Brownian motion are a.s. continuous (hence in $C([0,T])$). The important point here is that Brownian motion lives in a "nice" subset of the Hilbert space.

Measurability and the induced measure. The map β is measurable with respect to the Borel σ-algebra $\mathcal{B}\big(L^2([0,T])\big)$. We define the **Wiener measure** (or law of Brownian motion) as the push-forward:

$$\mu(A) \;:=\; P\big(\beta \in A\big) \;=\; P\big(\{\omega \in \Omega : \beta(\omega) \in A\}\big), \qquad A \in \mathcal{B}\big(L^2([0,T])\big).$$

Then μ is a **Gaussian measure** on $L^2([0,T])$ with **mean zero** (i.e. μ is centred).

Remark 38. Saying that μ is a Gaussian measure on $L^2([0,T])$ means that for every $h \in L^2([0,T])$, the real-valued random variable $\langle \beta,\, h\rangle_{L^2} = \int_0^T B_s\, h(s)\, \mu(ds)$ is normally distributed. This is indeed the case because $\int_0^T B_s\, h(s)\, ds$ is a Gaussian random variable for every deterministic $h \in L^2$ (as a limit of Riemann sums of jointly Gaussian random variables).

Verification that the mean is zero. Let $\{e_k\}$ be any CONS of $L^2([0,T])$. The k-th Fourier coefficient of a sample path is

$$x_k \;:=\; \langle B_\cdot,\, e_k\rangle_{L^2} \;=\; \int_0^T B_s\; e_k(s)\; ds.$$

Its expectation is

$$E(x_k) \;=\; E\Big(\int_0^T B_s\, e_k(s)\, ds\Big) \;=\; \int_0^T E(B_s)\, e_k(s)\, ds \qquad \text{(Fubini)}$$
$$= \int_0^T 0 \cdot e_k(s)\, ds \;=\; 0,$$

since $E(B_s) = 0$ for a standard Brownian motion. Because this holds for every k, the mean vector is $m = 0$.

Remark 39. The interchange of expectation and integral (application of **Fubini's theorem**) is justified because

$$E\Big(\int_0^T |B_s|\, |e_k(s)|\, ds\Big) \;\le\; \int_0^T E(|B_s|)\, |e_k(s)|\, ds \;\le\; \int_0^T \sqrt{s}\, |e_k(s)|\, ds \;<\; \infty,$$

using $E(|B_s|) = \sqrt{2s/\pi} \le \sqrt{s}$ and $e_k \in L^2([0,T])$.

Remark 40. One can further compute the **covariance operator** Q of the Wiener measure μ. For $h_1, h_2 \in L^2([0,T])$:

$$\langle Qh_1,\, h_2\rangle_{L^2} \;=\; E\Big(\int_0^T B_s\, h_1(s)\, ds \;\cdot\; \int_0^T B_r\, h_2(r)\, dr\Big) \;=\; \int_0^T\!\!\int_0^T \min(s,r)\; h_1(s)\, h_2(r)\; ds\, dr,$$

since $E(B_s B_r) = \min(s,r)$. Thus Q is the integral operator with kernel $K(s,r) = \min(s,r)$:

$$(Qh)(t) \;=\; \int_0^T \min(t,s)\; h(s)\; ds.$$

This operator is symmetric, positive semi-definite, and trace-class (with $\mathrm{tr}(Q) = \int_0^T t\, dt = T^2/2$), confirming that μ satisfies all the conditions of Theorem 1.6.

Detailed covariance computation. For $g, h \in L^2([0,T])$, the covariance of the Fourier coefficients $x_g := \langle \beta, g\rangle_{L^2}$ and $x_h := \langle \beta, h\rangle_{L^2}$ is computed as follows:

Remark 41. The following computation identifies the covariance operator Q of the Wiener measure by directly computing $\mathrm{Cov}(\langle \beta, g\rangle, \langle \beta, h\rangle)$ and recognising the result as an inner product with an integral operator. The key input is the covariance of Brownian motion, $E(B_s B_t) = \min(s,t)$, which encodes all the second-order information about the process.

$$\begin{aligned}
\mathrm{Cov}(x_g,\, x_h) \;&=\; E\big(\langle \beta, g\rangle\, \langle \beta, h\rangle\big) \\
&=\; E\Big(\int_0^T B_s\, g(s)\; ds \;\cdot\; \int_0^T B_t\, h(t)\; dt\Big) \\
&=\; E\Big(\int_0^T\!\!\int_0^T g(s)\, B_s\, B_t\, h(t)\; ds\, dt\Big) \qquad \text{(double integral)} \\
&=\; \int_0^T\!\!\int_0^T g(s)\; E(B_s\, B_t)\; h(t)\; ds\, dt \qquad \text{(Fubini)} \\
&=\; \int_0^T\!\!\int_0^T g(s)\; (s \wedge t)\; h(t)\; ds\, dt,
\end{aligned}$$

since $E(B_s B_t) = \min(s,t) = s \wedge t$.

Recognising the right-hand side as an L^2 inner product with the integral operator, we obtain:

$$\mathrm{Cov}(x_g,\, x_h) \;=\; \int_0^T g(s) \underbrace{\left(\int_0^T (s \wedge t)\, h(t)\, dt \right)}_{=(Qh)(s)} ds \;=\; \langle g,\, Qh \rangle_{L^2},$$

confirming that Q is the covariance operator with

$$(Qh)(s) \;=\; \int_0^T \min(s,t)\; h(t)\; dt.$$

Q **as the inverse of a differential operator.** We can split the integral defining Qg by separating the regions $s \le t$ and $s > t$:

$$\begin{aligned}(Qg)(t) \;&=\; \int_0^T \min(s,t)\; g(s)\; ds \\ &=\; \int_0^t g(s)\, s\; ds \;+\; t \int_t^T g(s)\; ds.\end{aligned}$$

Remark 42. In the first integral ($0 \le s \le t$), we have $\min(s,t) = s$. In the second integral ($t < s \le T$), we have $\min(s,t) = t$, which is pulled out of the integral as a factor.

Differentiating with respect to t using the **Leibniz rule**:

$$\begin{aligned}(Qg)'(t) \;&=\; \frac{d}{dt}\left[\int_0^t g(s)\, s\; ds \right] \;+\; \frac{d}{dt}\left[t \int_t^T g(s)\; ds \right] \\ &=\; g(t)\, t \;+\; \int_t^T g(s)\; ds \;-\; t\, g(t) \\ &=\; \int_t^T g(s)\; ds.\end{aligned}$$

Differentiating once more:

$$(Qg)''(t) \;=\; \frac{d}{dt}\left[\int_t^T g(s)\; ds \right] \;=\; -\, g(t).$$

Therefore:

$$\boxed{-(Qg)''(t) \;=\; g(t),}$$

which means Q is the **inverse of the negative second derivative** $-\partial_{tt}$, subject to the boundary conditions:

- **Dirichlet at** $t = 0$**:** $(Qg)(0) = \int_0^0 g(s)\, s\; ds + 0 \cdot \int_0^T g(s)\; ds = 0$.
- **Neumann at** $t = T$**:** $(Qg)'(T) = \int_T^T g(s)\; ds = 0$.

That is:

$$Q \;=\; \left(-\frac{d^2}{dt^2} \right)^{-1} \quad \text{on } L^2([0,T]), \quad \text{with Dirichlet BC at 0 and Neumann BC at } T.$$

Remark 43. This is a fundamental connection between probability and PDEs: the covariance operator of the Wiener measure is the **Green's operator** (inverse) of the negative Laplacian $-\partial_{tt}$ on $[0, T]$ with mixed boundary conditions. The kernel $K(s, t) = \min(s, t)$ is precisely the **Green's function** of this boundary value problem.

Concretely, solving the ODE $-u''(t) = g(t)$ on $[0, T]$ with $u(0) = 0$ and $u'(T) = 0$ gives $u(t) = \int_0^T \min(s, t)\, g(s)\, ds = (Qg)(t)$.

Remark 44. The boundary conditions have a natural probabilistic interpretation:

- The **Dirichlet condition** $(Qg)(0) = 0$ reflects the fact that Brownian motion starts at $B_0 = 0$ (no randomness at time 0).
- The **Neumann condition** $(Qg)'(T) = 0$ arises because Brownian motion has no "preferred direction" at the endpoint; its increments are symmetric.

Remark 45. This relationship $Q = (-\partial_{tt})^{-1}$ is an instance of a general principle: for many Gaussian processes, the covariance operator is the inverse of a differential operator. The **regularity** of the covariance kernel determines how "smooth" the typical sample paths are, and the order of the differential operator reflects this. Since $-\partial_{tt}$ is a second-order operator, its inverse Q is a "smoothing" operator of order 2, consistent with the fact that Brownian paths are Hölder-$\frac{1}{2}^-$ continuous (almost $\frac{1}{2}$ derivative of regularity).

Why the Laplace operator is not usually invertible

We have established that $Q = (-\Delta)^{-1}$, where Δ denotes the **Laplace operator** (here $\Delta = \frac{d^2}{dt^2}$ in one dimension).

However, the Laplace operator is **not usually invertible** on unbounded domains. Consider Δ on $\mathbb{R}$ acting on $L^2(\mathbb{R})$: the equation $-u''(t) = g(t)$ on $\mathbb{R}$ does not have a unique L^2 solution in general, because one can always add linear functions (which solve the homogeneous equation $u'' = 0$). Moreover, the spectrum of $-\Delta$ on $L^2(\mathbb{R})$ is $[0, \infty)$ (purely continuous, no gap at zero), so $-\Delta$ has no bounded inverse.

Remark 46. To make $-\Delta$ invertible, one needs to impose **boundary conditions** that pin down the solution. In our example, the **Dirichlet condition** $u(0) = 0$ (corresponding to the fixed starting point $B_0 = 0$ of Brownian motion) together with the Neumann condition at T is precisely what makes the operator invertible on the bounded interval $[0, T]$. On a bounded domain with appropriate boundary conditions, $-\Delta$ has a **discrete spectrum** $0 < \lambda_1 \leq \lambda_2 \leq \cdots$ and is invertible.

Why $Q = \mathrm{Id}$ is not a valid covariance operator

A natural first attempt might be to take the covariance operator to be the **identity** $Q = \mathrm{Id}$ (i.e., every direction has variance 1, as for a "standard Gaussian" in $\mathbb{R}^d$). However, this fails in infinite dimensions.

Trace condition. Let $\{e_k\}_{k=1}^{\infty}$ be any CONS of H. Then

$$\mathrm{tr}(\mathrm{Id}) \;=\; \sum_{k=1}^{\infty} \langle e_k,\, e_k \rangle_H \;=\; \sum_{k=1}^{\infty} 1 \;=\; +\infty.$$

Since the trace is **not finite**, $Q = \mathrm{Id}$ is *not* trace-class, and by Theorem 1.6 there is **no** Gaussian measure $\mathcal{N}(m, \mathrm{Id})$ on H.

Remark 47. This is a striking difference with finite dimensions. In $\mathbb{R}^d$, the standard Gaussian $\mathcal{N}(0,\, I_d)$ is perfectly well-defined because $\mathrm{tr}(I_d) = d < \infty$. In an infinite-dimensional Hilbert space H, there are **infinitely many independent directions**, each contributing variance 1, and the total variance $E(\|X\|_H^2) = \mathrm{tr}(Q) = \sum_k 1 = \infty$ diverges. Intuitively, the random vector would have infinite expected squared norm and would "not fit" inside H.

Geometric viewpoint: volumes of balls. There is a complementary geometric explanation. In $\mathbb{R}^d$, the volume of the unit ball B_1^d satisfies

$$\mathrm{Vol}(B_1^d) \;=\; \frac{\pi^{d/2}}{\Gamma(d/2+1)} \;\xrightarrow{d\to\infty}\; 0.$$

In infinite dimensions, interiors of balls have "zero volume": the Gaussian mass escapes to higher and higher dimensions. Formally, there is **no** σ-finite translation-invariant measure (analogue of Lebesgue measure) on an infinite-dimensional Hilbert space.

Remark 48. The fact that $\mathrm{Vol}(B_1^d) \to 0$ as $d \to \infty$ is a manifestation of the **curse of dimensionality**: in high dimensions, most of the "volume" of a Gaussian distribution is concentrated in a thin **annular shell** far from the origin, not near the centre. For a valid Gaussian measure on H, the covariance operator Q must have **eigenvalues** $\lambda_k \to 0$ fast enough that $\sum_k \lambda_k = \mathrm{tr}(Q) < \infty$. This forces the Gaussian to "shrink" in most directions, compensating for having infinitely many of them.

Remark 49. For the Wiener measure, the eigenvalues of $Q = (-\Delta)^{-1}$ on $[0, T]$ with Dirichlet–Neumann boundary conditions are $\lambda_k = \frac{1}{\mu_k}$, where μ_k are the eigenvalues of $-\Delta$. Since $\mu_k \sim k^2$ for large k, the eigenvalues $\lambda_k \sim k^{-2}$ decay to zero, and $\mathrm{tr}(Q) = \sum_k \lambda_k < \infty$ (the series $\sum k^{-2}$ converges), confirming that the Wiener measure is a well-defined Gaussian measure on $L^2([0, T])$.